The triangular lattice is not a local minimum of the harmonic zero-point energy for $e^{-\pi r^2}$: a certified linear-response majorant along a symmetric shear

Xavier Callens (SocrateAI Lab)

Preprint v1, 2026-10-26

Zenodo DOI to be assigned on release (not peer reviewed)

AI-assistance disclosure. This preprint, its Lean code, its Python code and its claim ledger were produced by AI agents (Claude, Anthropic) under the author's direction. The author is responsible for the content. No human has audited the Lean statements for adequacy, and no human referee has checked the hand arguments. An independent adversarial audit by a separate AI agent instance (`AUDIT.md`, verdict ACCEPT_WITH_CORRECTIONS) recompiled the Lean files, re-ran the ball arithmetic with coarser cells and re-implemented the quantities in floating point. The main theorem statement is locked in `results/proposal2/2026-10-26/B2/STATEMENT_LOCK.json`. No human has audited the work.

Abstract

For the Gaussian pair potential $V(r) = e^{-\pi r^2}$ and unit-covolume planar Bravais lattices, let $E(t)$ be the harmonic zero-point energy per particle ($\hbar = m = 1$) along the symmetric shear $L_t = P_t L_\rho$, $P_t = \left(\begin{smallmatrix} 1 & t \\ t & 1+t^2 \end{smallmatrix}\right)$, starting at the triangular lattice L_ρ. We show that $E(t) < E(0)$ for every $0 < |t| \le 10^{-5}$, so **the triangular lattice is not a local minimum of the zero-point energy**. The proof bounds E above by a majorant U that is affine in the dynamical matrix (a tangent form of the concavity of $\mathrm{tr}\sqrt{\cdot}$) and certifies that the second difference of U is at most $-1.84156\,t^2$. This is an *upper bound* on a curvature, not a two-sided enclosure of the Laplacian of E_{zp} at ρ.

Tier split. [A]: the Lean 4.34.1 kernel checks the 2×2 concavity inequality, the logic (a majorant window bound with $B < 0$ excludes a local minimum), the rational certificate ($\mathrm{Upp0}_{hi} + \mathrm{trW}_{hi}\,\mathrm{esum}_{hi} < 0$, stability margin > 0) and the assembled theorem, whose analytic inputs are explicit hypotheses; axioms stay within {`propext`, `Classical.choice`, `Quot.sound`}, and planted-`sorry`, planted-axiom, tampered-certificate and $a = 2$ controls are caught or rejected. [C-ball]: the Brillouin-zone integrals, lattice sums and tails are rigorous Arb ball-arithmetic enclosures (206,468 cells, 0 failed) executed by python-flint, *not kernel-checked*. [C-human]: the glue ($E \le U$ after integration, the window identity, mirror symmetry, identification with the earlier E_{zp}). Hence the headline is Tier C overall; only its logic and arithmetic are Tier A. A corollary $\mathrm{Lap}\,E_{\mathrm{zp}}(\rho) \le -0.92078$ needs extra hypotheses ($E \in C^2$ and E_{zp} of class C^2 near ρ; ρ critical and the Hessian isotropic, both by the order-3 stabilizer, P2 3.1) and is [C-human]. The same certificate also works at $a = 1/2$ (not preregistered), and fails, as it must, at $a = 2$.

Tiers and markers. [A]: kernel-checked Lean, axioms within the allowlist, controls caught. [L]: a cited, quoted source (none carries a claim here; the literature in the bibliography is context). [C-ball]: rigorous ball arithmetic, outward rounded, executed but not kernel-checked. [C-human]: a written mathematical argument, not machine-checked. [X]: floating point, supporting nothing by itself. Identifiers `CZP-·-nnnn` refer to `ledger/ledger.json` (Section 6); in the ledger both C flavours are Tier C and the flavour is the `subtier` field of the evidence blob. A Tier A tag attaches to the Lean statement, never to its physical gloss.

1 Setting, and what was preregistered

This is a follow-up to the programme paper on the zero-point energy [1] ("P2"), which found numerically (Tier X) that the triangular lattice is *not* the minimiser of E_{zp} for several completely monotone potentials, and

that the curvature of E_{zp} at $\rho = e^{i\pi/3}$ is negative for Gaussians $e^{-\pi a r^2}$ with a in an interval containing $a = 1$ (P2 value $\operatorname{Lap} E_{\mathrm{zp}}(\rho) = -3.848$ at $a = 1$, Tier X). The present work asks for a *certified* statement at $a = 1$.

The functional. Let $L_\rho = c\mathbb{Z}(1,0) + c\mathbb{Z}(\frac{1}{2}, \frac{\sqrt{3}}{2})$, $c^2 = 2/\sqrt{3}$, and $R_n(t) = P_t R_n(0)$ for integer labels $n = (m, l)$. Put $H_n(t) = \operatorname{Hess} V(R_n(t)) = e^{-\pi|R|^2}(4\pi^2 R R^{\mathsf{T}} - 2\pi I)|_{R=R_n(t)}$ and, at reduced zone coordinate $\kappa \in [-\frac{1}{2}, \frac{1}{2}]^2$,

$$D_t(\kappa) = \sum_{n \neq 0} \bigl(1 - \cos 2\pi\kappa\cdot n\bigr) H_n(t), \qquad E(t) = \tfrac{1}{2} \int \operatorname{tr} D_t(\kappa)^{1/2}\, d\kappa.$$

The labels n and κ are held fixed as t varies (strain frame, no rotation). $\det P_t = 1$, so the covolume stays one, and the lattice shape is $\tau(t) = \rho + \sqrt{3}\, t + O(t^2)$. [C-human] (`CZP-C-0006`)

Preregistration and deviation. Item B2 of `results/proposal2/2026-10-26/preregistration.json` (prior 0.4) reads: "the isotropic curvature $\operatorname{Lap} E_{\mathrm{zp}}$ at the triangular point is negative for $a = 1$, certified by interval arithmetic with rigorous tail bounds and checked in Lean; negative control: $a = 2$ is not certified negative"; kill rule: "if the rigorous enclosure cannot exclude 0, report the best certified interval". **What was delivered differs.** A direct enclosure of $\operatorname{Lap} E_{\mathrm{zp}}$ was not attempted; the producer chose the linear-response majorant because it is affine in D_t and needs no matrix square root of the perturbed lattice. (The second variation of $\operatorname{tr} D^{1/2}$ is smooth away from Γ; a direct certified enclosure of $E''(0)$ with the same Duffy machinery appears feasible and is left open.) Instead the method certifies a majorant, which yields "not a local minimum" (the title) without isotropy or criticality, and a one-sided Laplacian bound only as a [C-human] corollary with extra hypotheses. The preregistered literal statement (a two-sided certified $\operatorname{Lap} E_{\mathrm{zp}} < 0$, checked in Lean) is therefore **not** established; what is established is weaker in one respect (one-sided, local, one path) and different in another (no isotropy needed). The negative control ($a = 2$ not certified) was run as preregistered.

2 Step-by-step human demonstration, with the matching Lean statements

Each step states the mathematics, then the Lean counterpart (full text in `data/lean/Basic.lean`, hash in Section 5). Listings are transliterated to ASCII.

Step 1: a tangent form of the concavity of $\operatorname{tr}\sqrt{\cdot}$ [A] (`CZP-A-0001`). Let X be symmetric positive definite, S symmetric, $P = X^{-1}$. Then

$$2\operatorname{tr} S \leq \operatorname{tr} X + \operatorname{tr}(PS^2), \qquad \text{with equality at } S = X.$$

Proof. Put $M = S - X$. Expanding, $\operatorname{tr}(P(S-X)^2) = \operatorname{tr}(PS^2) - \operatorname{tr}(PSX) - \operatorname{tr}(PXS) + \operatorname{tr}(PX^2) = \operatorname{tr}(PS^2) - 2\operatorname{tr} S + \operatorname{tr} X$, using $PX = I$ and $\operatorname{tr}(PSX) = \operatorname{tr}(XPS) = \operatorname{tr} S$. Since M is symmetric, $\operatorname{tr}(PM^2) = \sum_j m_j^{\mathsf{T}} P m_j \geq 0$ over the columns m_j of M, and $P \succ 0$. □ In Lean the 2×2 case is a `linear_combination` identity (the displayed trace identity, using the four entries of $XP = I$) plus two uses of `quad_nonneg` (one per column).

```
theorem quad_nonneg (p11 p12 p22 a b : R) (hp : 0 < p11) (hdet : 0 <= p11*p22 - p12^2) :
    0 <= p11*a^2 + 2*p12*a*b + p22*b^2
theorem two_trace_le_of_inverse (x11 x12 x22 p11 p12 p22 s11 s12 s22 : R) (hp : 0 < p11)
    (hdet : 0 <= p11*p22 - p12^2) (h11 : x11*p11 + x12*p12 = 1) (h12 : x11*p12 + x12*p22 = 0)
    (h21 : x12*p11 + x22*p12 = 0) (h22 : x12*p12 + x22*p22 = 1) :
    2*(s11+s22) <= (x11+x22) + (p11*(s11^2+s12^2) + 2*p12*(s12*(s11+s22)) + p22*(s12^2+s22^2))
```

The right side is $\operatorname{tr} X + \operatorname{tr}(PS^2)$ written out for symmetric S. The statement is for 2×2 matrices only, which is all that is needed (two phonon branches). `two_trace_eq_of_eq` is the equality case.

Step 2: the majorant U [C-human] (`CZP-C-0007`). At each $\kappa \neq 0$ take $S = D_t(\kappa)^{1/2}$ (so $S^2 = D_t$) and $X = D_0(\kappa)^{1/2}$, positive definite because L_ρ is dynamically stable. Step 1 gives

$$\operatorname{tr} D_t^{1/2} \leq \tfrac{1}{2}\bigl(\operatorname{tr} X + \operatorname{tr}(X^{-1} D_t)\bigr).$$

Integrate: $E \le U$ where $U(t) = \frac{1}{4}\int[\operatorname{tr} X + \operatorname{tr} X^{-1}D_t]\,d\kappa$, and $U(0) = E(0)$ by the equality case. The integrand is integrable because $\operatorname{tr}(X^{-1}D_t) \le \|D_t\| \operatorname{tr} X^{-1}$ and $\int \operatorname{tr} X^{-1} \le 1.8336$ ([C-ball], Step 5). *Why a majorant: U* is *affine* in D_t, so its dependence on t is a single absolutely convergent lattice sum against the fixed matrix X^{-1}, with no square root and no branch degeneracy. The price is that U only bounds E from above; the neglected term $\operatorname{tr}(X^{-1}(S-X)^2) \ge 0$ can only help the conclusion "E is smaller". Lean sees only the pointwise inequality; the integration, integrability and the identification $S = D_t^{1/2}$ are glue. (Because the majorant is built in the strain frame, it is lower-curvature than P2's tau-frame linear response; this frame dependence is expected and is why the number below differs from P2's decomposition.)

Step 3: the window identity [C-human] (`CZP-C-0008`) with path derivatives [A] (`CZP-A-0006`). Since U is affine in D_t,

$$U(t) + U(-t) - 2U(0) = \tfrac{1}{4}\int \operatorname{tr}\Big(X^{-1}\sum_n (1-\cos 2\pi\kappa\cdot n)\big[H_n(t) + H_n(-t) - 2H_n(0)\big]\Big)\,d\kappa.$$

Entrywise, the second-order mean value theorem gives $H_n(t) + H_n(-t) - 2H_n(0) = t^2 H_n''(\xi)$ with $|\xi| < t$. Write $H_n''(\xi) = H_n''(0) + (H_n''(\xi) - H_n''(0))$. With $G = \sum_n (1-\cos)H_n''(0)$, $\mathrm{Upp0} = \frac{1}{4}\int \operatorname{tr}(X^{-1}G)\,d\kappa$ $(= U''(0))$, and using $\|W\|_{op} \le \sum |W_{ij}|$ and $1 - \cos \le 2$ (which, with the prefactors, leaves a slack in the code's constant),

$$\frac{U(t) + U(-t) - 2U(0)}{t^2} \le \mathrm{Upp0} + \Big(\int \operatorname{tr} X^{-1}\Big)\cdot \mathrm{esum}, \quad 0 < t \le t_0 = 10^{-5},$$

where $\mathrm{esum} = \sum_n \sum_{\text{entries}} \sup_{|\xi|\le t_0} |H_n''(\xi) - H_n''(0)|$ is a sum over *all* lattice points (tail included). The Lean counterpart that *is* kernel-checked is the path calculus: $q(t) = |P_t R_0|^2$, q', q'' (`hasDerivAt_qS`, `hasDerivAt_q1S`), the Gaussian weight $e = e^{-\pi a q}$ with $e' = -\pi a q' e$ and $e'' = e(\pi^2 a^2 q'^2 - \pi a q'')$ (`hasDerivAt_gaussS`, `hasDerivAt_gaussS'`, reusing P2's `gauss_f'`), and the double product rule for one Hessian entry $H = e\,M$:

```
theorem hasDerivAt_entry' {e e1 M M1 : R -> R} {e2 M2 t : R} (he1 : HasDerivAt e1 e2 t)
    (hM : HasDerivAt M (M1 t) t) (he : HasDerivAt e (e1 t) t) (hM1 : HasDerivAt M1 M2 t) :
    HasDerivAt (fun u => e1 u * M u + e u * M1 u) (e2 * M t + 2 * e1 t * M1 t + e t * M2) t
```

Assembling these into the entry formula that the code evaluates ($M = 4\pi^2 a^2 R_i R_j - 2\pi a \delta_{ij}$ along P_t) is [C-human] (`CZP-C-0009`); a finite-difference check agrees to 7×10^{-7} [X].

Step 4: mirror symmetry [C-human] (`CZP-C-0010`). With $\sigma = \mathrm{diag}(1,-1)$: $\sigma L_\rho = L_\rho$ because $\sigma\, c(\frac{1}{2}, \frac{\sqrt{3}}{2}) = c(1,0) - c(\frac{1}{2}, \frac{\sqrt{3}}{2})$, and $\sigma P_t \sigma = P_{-t}$, so $L_{-t} = \sigma L_t$; E depends on the lattice only up to isometry, so $E(-t) = E(t)$. Tier X check: $E(h) - E(-h) = 0.0$ to all printed digits.

Step 5: the numbers [C-ball] (`CZP-C-0001, -0002`) and their arithmetic [A] (`CZP-A-0003`). Ball arithmetic encloses $\mathrm{Upp0} = U''(0)$, $\int \operatorname{tr} X^{-1}$, esum and a stability margin (Section 3). Their rational upper endpoints are copied into Lean:

```
def Upp0_hi : R := -475041875131 / 250000000000     def trW_hi : R := 366706720019 / 200000000000
def esum_hi : R := 31964165579 / 1000000000000      def t0 : R := 1 / 100000
theorem cert_B_neg : Upp0_hi + trW_hi * esum_hi < 0
theorem cert_stability : 0 < lmin_lo - dB_hi
theorem cert_window (Upp0 trW esum : R) (hU : Upp0 <= Upp0_hi) (hW0 : 0 <= trW) (hW : trW <= trW_hi)
    (he0 : 0 <= esum) (he : esum <= esum_hi) : Upp0 + trW*esum <= Upp0_hi + trW_hi*esum_hi
```

The kernel proves the sign and the monotonicity; whether those endpoints really bound the integrals is the [C-ball] content, and that the literals equal the run file is checked by a script (`lean_literal_provenance`, all match). LeanFlow `lean.certify` independently renders $\mathrm{Upp0}_{hi} + \mathrm{pert}_{hi} < 0$ and $\mathrm{lmin}_{lo} - \mathrm{dB}_{hi} > 0$ into integer statements and verifies both ([A], `CZP-A-0004`).

Step 6: the logic [A] (`CZP-A-0002`). Suppose $E \le U$ on $|t| \le t_0$, $U(0) = E(0)$ and $U(t) + U(-t) - 2U(0) \le t^2 B$ with $B < 0$. Then for $0 < t \le t_0$: $E(t) + E(-t) \le U(t) + U(-t) \le 2U(0) + t^2 B < 2E(0)$. So at least one of $E(t), E(-t)$ is $< E(0)$ for every small t, hence 0 is not a local minimum; with $E(-t) = E(t)$ (Step 4), both are.

```
theorem not_isLocalMin_of_majorant {E U : R -> R} {t0 B : R} (ht0 : 0 < t0) (hB : B < 0)
    (hEU : forall t, |t| <= t0 -> E t <= U t) (h0 : U 0 = E 0)
    (hU : forall t, 0 < t -> t <= t0 -> U t + U (-t) - 2*U 0 <= t^2 * B) : Not (IsLocalMin E 0)
theorem lt_of_majorant_window_symm ... (hsym : forall t, E (-t) = E t) (ht : 0 < t) (htt : t <= t0) : E t < E
    0
```

Step 7: assembly [A] (conditional, `CZP-A-0005`) and headline [C] (`CZP-C-0012`). `triangular_not_localMin_of_certificate` takes $E \le U$, $U(0) = E(0)$, the window bound with $\mathrm{Upp0}, \mathrm{trW}, \mathrm{esum}$, their upper bounds, and $E(-t) = E(t)$ as hypotheses and concludes $\neg\,\mathrm{IsLocalMin}\,E\,0 \;\wedge\; \forall t \in (0, t_0],\, E(t) < E(0)$. Every hypothesis is a [C-ball] or [C-human] fact, so the Lean theorem is a precise statement of what the analytic work must supply, not a proof of the physics. `cert_stability` is proved but is not a hypothesis of `triangular_not_localMin_of_certificate`; it supports the [C-human] discharge of the hypothesis $E \le U$ (D_t must be positive semidefinite for $S = D_t^{1/2}$). The hypothesis $E(-t) = E(t)$ is stated for all t although $|t| \le t_0$ suffices; it is true globally, so nothing depends on this.

Step 8: why this excludes a local minimum of the lattice-shape problem [C-human] (`CZP-C-0011`). Because $\tau(t) = \rho + \sqrt{3}\,t + O(t^2)$, small $t \neq 0$ give unit-covolume lattices of shape $\neq \rho \pmod{SL_2(\mathbb{Z})}$ arbitrarily close to ρ, and each has strictly smaller E_{zp} than L_ρ. A local minimum of E_{zp} over shapes at ρ would require $E_{\mathrm{zp}} \ge E_{\mathrm{zp}}(\rho)$ on a shape neighbourhood, contradicted along this path. No criticality or isotropy at ρ is used. (Strict $E(t) < E(0)$ already implies a different shape; the τ-speed computation is needed only for the corollary of Step 9.)

Step 9 (corollary, extra hypotheses) [C-human] (`CZP-C-0013`). If moreover E is C^2 at 0, the window bound gives $E''(0) \le B_{hi} = -1.84156$. If E_{zp} is of class C^2 near ρ (needed for the chain rule), ρ is critical ($\nabla E_{\mathrm{zp}}(\rho) = 0$, otherwise a $\nabla E_{\mathrm{zp}} \cdot \tau''(0)$ term appears; this follows from the order-3 stabilizer of ρ in $SL_2(\mathbb{Z})$), and the curvature of E_{zp} at ρ is isotropic (P2 section 3.1, itself Tier C; also from the order-3 stabilizer) and $|\tau'(0)|^2/y^2 = 4$ (so $E''(0) = 2\,\mathrm{Lap}\,E_{\mathrm{zp}}(\rho)$), then $\mathrm{Lap}\,E_{\mathrm{zp}}(\rho) \le -0.92078 < 0$. This is consistent with, and far weaker than, P2's Tier X value -3.848; Tier X here gives $E''(0) = -7.696$.

3 Certified numbers and what they mean

All values are outward-rounded ball endpoints (Arb, python-flint 0.6.0, 96 bits) for $a = 1$, shear path, $t_0 = 1/100000$ (file `data/run_a1_shear_main.json`); [C-ball].

quantity	enclosure / bound	meaning
$U''(0) = \mathrm{Upp0}$	$[-1.947713, -1.900168]$	second derivative of the majorant (tier X quadrature -1.923973 inside)
$\int \mathrm{tr}\, X^{-1}\, d\kappa$	$[1.825520, 1.833534]$	weight multiplying the perturbation
esum	≤ 0.0319642	all-lattice bound on $\sup\lvert H''(\xi) - H''(0)\rvert$, tails included
$B_{hi} = \mathrm{Upp0}_{hi} + \mathrm{trW}_{hi}\,\mathrm{esum}_{hi}$	-1.841560	**upper bound** on $(U(t) + U(-t) - 2U(0))/t^2$
$\lambda_{\min}(B_0)$	≥ 0.918889	stability of L_ρ on every cell
$\lVert B_t - B_0 \rVert$	≤ 0.007528	stability margin $0.911361 > 0$ for $\lvert t\rvert \le 10^{-5}$

As rationals, $\mathrm{Upp0}_{hi} = -475041875131/250000000000$, $\mathrm{trW}_{hi} = 366706720019/200000000000$, $\mathrm{esum}_{hi} = 31964165579/10^{12}$ (these appear verbatim in the Lean file). *Meaning.* B_{hi} is an upper bound on a curvature of the majorant U, which in turn bounds a second difference of E from above. It is **not** a two-sided enclosure of $\mathrm{Lap}\,E_{\mathrm{zp}}(\rho)$ and cannot certify a local minimum; the method bounds $E'' \le U''$ from one side only. Engine details ([C-ball]): the reduced square is split into four triangles with apex Γ and Duffy coordinates $\kappa = s\,v(t')$, Jacobian $s/2$; with $D_0 = s^2 B$ the integrands times the Jacobian are C^1 in (s, t') wherever $B > 0$, including $s = 0$. Each cell uses the mean-value form $\int_{\mathrm{cell}} g \in |\mathrm{cell}|\, g(\mathrm{centre}) + |\mathrm{cell}|\,\frac{h}{4}(\sup|g_s| + \sup|g_{t'}|)[-1, 1]$; 64×64 cells per triangle, bisected up to depth 3 to 206,468 cells, 0 failed, 448 s on one core. The lattice sum is exact over $m^2 + ml + l^2 \le 21$ (42 pairs $\pm n$) and the tails ($|R| \ge 5.04$) use the lattice-point count $\#\{|R| \le r\} \le \pi(r+1)^2$

and Gaussian-times-polynomial bounds, below 10^{-21} per entry. The remainder beyond the 200 summed shells is bounded in the code by twice the last shell; the decay ratio below 1/2 behind this is *asserted, not computed*: at $r \approx 55$ with shell width 1/4 and $a \geq 1/2$ it is below e^{-40} by the audit's estimate. On coarse grids the certifier fails loudly rather than returning a wrong ball: with $N = 16$ cells per triangle side it raises `fprime_enc: ball too wide` (the independent audit hit this at $N = 16$, depth 2), so it cannot run below roughly $N = 32$ without splitting straddling balls. The esum tail contained a factor-2 undercount (`3*bh2` instead of `6*bh2`); it is corrected in the code files here, and its effect is nil (tail 2.2×10^{-25} against 9.2×10^{-13} of outward rounding already added to `esum_hi`); the certification was *not* re-run, and the reported runs used the earlier file (sha256 below). These bounds are mathematics checked only by being coded; they are not kernel-checked, and a bug in them would not be caught by the Lean theorems.

4 Controls

control	what was tested / result	tier
(+)1 $E(0)$ enclosure	$[1.498719, 1.507934]$ contains P2's spectral 1.50335235220736	[C-ball] (`C-0003`)
(+)2 $\langle \mathrm{tr}\, D \rangle$	cell value $[5.26378, 5.29693]$ contains the exact lattice sum 5.2804187	[C-ball]
(+)3 sum rule	5.2804187 vs $4\pi - 2\pi\theta_L(1)$, $\theta_L(1) = 1.1595952670$ (ball): overlap	[C-ball]
(+)4 second path $\mathrm{diag}(1{+}t, \frac{1}{1+t})$	$U''(0) \in [-1.945739, -1.902147]$, overlaps the shear path; $B_{hi} = -1.83885$	[C-ball]
(+)5 Hessian formula	closed form vs finite differences, max difference 7×10^{-7}	[X]
(+)6 trace identity	sympy identity behind Step 1 is 0; a perturbed version is not	[X]
(−)1 $a = 2$ (triangular is a local min, P2)	$U''(0) \in [+12.69365, +12.75701]$, $B \in [12.772, 12.836]$: **not certified**. LeanFlow: rejected (`false-statement`); `RejectA2.lean`: kernel rc 1, goal `False`; `a2_not_certifiable` proves no trW, esum ≥ 0 can rescue it. Stability margin $0.1452 > 0$	[C-ball] + [A] (`C-0005`, `A-0007`, `A-0008`)
(−)2 tampered certificate (sign of $\mathrm{Upp0}_{hi}$ flipped)	`RejectTampered.lean` rc 1 (goal `False`); `tampered_false` proves the negation	[A]
(−)3 planted `sorry` / planted axiom proving the tampered claim	`Controls.lean` compiles with rc 0, but `#print axioms` shows `sorryAx` and `ProgZeroPointCert.ctl_cheat` (the exit code is not the gate)	[A] (`A-0008`)

The $a = 2$ run is the method's falsification test: at $a = 2$ P2 finds the triangular lattice *is* a local minimum, and the majorant correctly yields a positive bound, so the certificate is not produced. The test shows that the method does not always say "not a minimum"; it does not show the method is complete.

Extra result at $a = 1/2$ (NOT preregistered, reported as a by-product) [C-ball] (`CZP-C-0004`). Same code, same path: $U''(0) \in [-1.043919, -0.982646]$, $B_{hi} = -0.918357 < 0$, stability margin 0.3648; LeanFlow tier A verified the endpoint arithmetic. So the same certificate covers $a = 1/2$ (P2 Tier X: triangular is a local *maximum* there). It is *not* part of the Lean `Basic.lean` theorem, which carries the $a = 1$ literals only; its $E(0)$ enclosure $[1.503411, 1.509136]$ contains P2's 1.50628947. A Tier C interpretation, not tested by a separate run: P2's linear-response term is $+0.028 > 0$ at $a = 1/2$ in the rotation-inclusive τ frame, while the strain-frame majorant here omits the rotation direction (along which U has a minimum, since E is rotation invariant and $U \geq E$); the frame dependence of U is expected.

5 Lean formalization and LeanFlow

Module. `formal_cert/Programmes/ZeroPointCert/Basic.lean` (module `Programmes.ZeroPointCert.Basic`, namespace `ProgZeroPointCert`); it imports `Programmes.ZeroPoint.Basic` and Mathlib submodules, never `import Mathlib`, and contains no `sorry`. Compiled with the pinned Lean 4.34.1 stack through `scripts/certified_numerics/lean_pinned.py`: rc 0, 0 errors, 33.4 s. All 18 printed declarations depend on exactly {`propext`, `Classical.choice`, `Quot.sound`} (`data/lean_basic.json`): `quad_nonneg`, `two_trace_le_of_inverse`, `two_trace_eq_of_eq`, `sum_lt_of_majorant_window`, `not_isLocalMin_of_majorant`, `lt_of_majorant_window_symm`, `t0_pos`, `cert_B_neg`, `cert_stability`, `cert_window`, `triangular_not_localMin_of_certificate`, `hasDerivAt_lin`, `hasDerivAt_quad`, `hasDerivAt_qS`, `hasDerivAt_q1S`, `hasDerivAt_gaussS`, `hasDerivAt_gaussS'`, `hasDerivAt_entry'`.

Hashes (sha256).

- `Basic.lean`: 0241b6afa500e46a2fb79cdedc7770eb20cdb0dc336283d683d55096f4d54c00
- `Controls.lean`: 2e95e21b70af3b4a1719b761636c992cfec8d52a61a7c63962f02ff5562d4ff3
- `RejectTampered.lean`: 02f6930b22e47d7cdc26c89a13102bb1c7af33351e80ec09b7aa4b7ef22e4288
- `RejectA2.lean`: ab0f5f297b0a24bafa84175d66e225fbe626a941bbcff31c28dd7010c28dada9

Controls in Lean. `Controls.lean` (rc 0) contains the planted `ctl_sorry` and `ctl_axiom`; `#print axioms` reports `sorryAx` and `ProgZeroPointCert.ctl_cheat` respectively, so a gate that read only the exit code would accept them. The two reject files (rc 1 each) are false inequalities that `norm_num` cannot close.

LeanFlow. The unmodified `lean.certify` tool of LeanFlow (checkout `/home/callensxavier_gmail_com/LeanFlow-wt-p1`) renders the claim files `leanflow_claim_a1.json` etc. into integer statements discharged by `decide`. Results: $a = 1$ *verified* (status ok, tier A for the arithmetic statement only, empty axiom lists); $a = 1/2$ verified; $a = 2$ *rejected* (`false-statement`, `cert_0`). LeanFlow's own scope note applies: it certifies the arithmetic statement about the stated numbers, not their provenance. Here $\mathrm{pert}_{hi} = \mathrm{trW}_{hi}\mathrm{esum}_{hi}$ is formed exactly in Python fractions for LeanFlow, while the kernel redoes the product in `cert_B_neg`.

What Tier A certifies. The 2×2 concavity inequality, the window logic, the rational sign and monotonicity, the path derivatives and the conditional assembly. *It does not certify* that E and U are the stated integrals, any enclosure, the identification with E_{zp} of P2, or any statement about Laplacians.

6 Claim ledger

`ledger/build_ledger.py` files 23 claims: 8 at Tier A (`CZP-A-0001`–`0008`), 13 at C (5 [C-ball]: `C-0001`–`0005`; 8 [C-human]/headline: `C-0006`–`0013`) and 2 at X. Each has a content-addressed evidence blob in `ledger/evidence/`. The tier discipline of the gate forces the headline `CZP-C-0012` to Tier C because it rests on C-ball and C-human rows; the Lean rows stay at A. The gate (`tools/ledger.py ledger.json --evidence-dir evidence`; verbatim output in `ledger/gate_output.txt`) exits with code **1**: **0 blocking findings and 8 flags** (`LEDGER_UNAUDITED_TIER_A`, one for each Tier A row; the gate's wording is that no *person* has certified that the Lean statements mean what their glosses say). The independent AI audit (`papers/p2_certified_zero_point/AUDIT.md`, verdict ACCEPT_WITH_CORRECTIONS) and the statement lock (`results/proposal2/2026-10-26/B2/STATEMENT_LOCK.json`, source hash `d6947d20...`) are recorded for `CZP-A-0001`–`0008` in a separate ledger field `ai_audit`. The gate's `audit` field is left null on purpose: the gate requires a human audit to clear `LEDGER_UNAUDITED_TIER_A`, and an AI audit is not one. The 8 flags therefore remain open, and the gate output quoted in `ledger/gate_output.txt` is unchanged in substance.

7 Limitations

- **Mostly Tier C.** Only the logic and the rational arithmetic are kernel-checked. The ball-arithmetic enclosures (the cell bounds, Duffy map, tails, lattice count) are executed code, not proofs; the glue steps (Steps 2–4, 8) are written arguments checked by no machine and no human referee.

- **Window and scope.** $|t| \leq 10^{-5}$, one path, $a = 1$, Bravais lattices, harmonic approximation, two dimensions. The theorem is local; it says nothing about the finite competitor $0.5 + 0.966i$ of P2, which remains Tier X.
- **One-sided.** The method cannot certify a local minimum, and it gives an upper bound on curvature only. The preregistered two-sided certified Laplacian was not obtained (Section 1).
- **Statement locked and independently audited by an AI agent (not a human); the audit's corrections are applied; the Tier A adequacy flags are closed by that audit only to the extent recorded in the ledger's audit fields.** In the ledger the AI audit is recorded in `ai_audit`; the gate's `audit` field stays null, so the eight Tier A flags remain open. No human has audited the work.
- **Repository guards.** `python antigravity_guard.py` and `python test_rigor_guard.py` were run at the time of the B2 result and *both fail* for reasons that predate this work: hollow tests in `tests/test_k3_anse_improvements.py` and `tests/test_scientific_rigor_v14.py`, and hallucinated-import reports for packages (e.g. `peft`, `trl`, `fastmcp`, `hypothesis`) not installed in the venv. Verbatim outputs: `data/guard_antigravity.txt`, `data/guard_test_rigor.txt`. They were not repaired here, and "guards pass" is not claimed.
- **python-flint location.** python-flint 0.6.0 is not in the project venv; it was installed with `pip --target` into `.scratchpad/b2_pylib` (git-ignored scratch) and put on `sys.path` by `b2_certify.py`. Re-running requires reinstalling that version.
- **Missing main-run log.** `run_a1_shear_main.json` exists, but the plain-text log of the main $a = 1$ run was not archived (logs exist for the diagonal, $a = 2$ and $a = 1/2$ runs). The *b2_explore2.py* exploratory output behind the Tier X cross-checks is likewise not archived; those numbers are quoted from the result note.
- **Not novelty-checked.** No literature search was made for prior certified results on zero-point lattice energies; no novelty is claimed.

8 Reproducibility

Repository AutoevolveAI, branch `worktree-openai-math-discovery`; B2 files committed in `b687db2` (Lean, results); HEAD when this paper was written: `74b908125942398600f49e72d7da5b177acdbed9`. Nothing was pushed or published. Copies of the inputs are in `data/` (`data/code/` holds the scripts). Commands (cores 6,7; from the repository root; a is the Gaussian parameter):

```
python -I code/b2_explore2.py 1 64                                  # Tier X cross-checks
B2_REFINE_TOL=0.01 B2_MAX_DEPTH=3 python -I code/b2_certify.py 1 shear 64 1/100000 21 run_a1_shear_main.json   # 448 s
B2_REFINE_TOL=0.01 B2_MAX_DEPTH=3 python -I code/b2_certify.py 1 diag 64 1/100000 21 run_a1_diag_ctl.json          # diagonal-
    path control
B2_REFINE_TOL=0.01 B2_MAX_DEPTH=3 python -I code/b2_certify.py 2 shear 64 1/100000 21 run_a2_shear_negctl.json # negative
    control
B2_REFINE_TOL=0.01 B2_MAX_DEPTH=3 python -I code/b2_certify.py 1/2 shear 64 1/100000 21 run_a05_shear_limit.json
python -I code/b2_lincomb.py ; python -I code/b2_tau_speed.py
python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/ZeroPointCert/Basic.lean --olean Programmes.
    ZeroPointCert.Basic
python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/ZeroPointCert/Controls.lean        # rc 0; #
    print axioms shows sorryAx / ctl_cheat
python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/ZeroPointCert/RejectTampered.lean   # must
    fail
python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/ZeroPointCert/RejectA2.lean        # must
    fail
python -I ledger/build_ledger.py ; python -I ledger/run_gate.py     # gate: exit 1 = 0 blocking, 8 flags
```

Script hash of the certifier used for all reported runs: `b2_certify.py` sha256 `168115eb63b6c97728fc3dd3b586200c05bb5ebfba25a6a2da640f1973d3338c`. After the audit (NB-1, NB-6) the file was edited (esum tail factor and comments only); the current file has sha256 `d8a97f8826a8b41b76b6a0f56716222c9ee3aa71b2efb816f75d09ba45939df8`, and the run files still record the earlier hash. Not re-run, since the effect is nil. The Lean files are unchanged (hashes above). The long runs take 7 to 22 minutes each.

References

[1] X. Callens, *Does the triangular lattice minimise the harmonic zero-point energy? A numerical counterexample for completely monotone potentials, with kernel-checked curvature identities*, Zenodo, doi:10.5281/zenodo.23653833 (2026). Own preprint (the tier-X values -3.848, 1.50335235220736, and the isotropy argument are taken from it); not peer reviewed.

[2] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, arXiv:1611.02831 (2016); the page was fetched on 2026-10-26 to check title, author and year; its journal version (IEEE Trans. Comput. 66(8), 2017) was not opened and is not cited by page.

[3] The mathlib Community, *The Lean mathematical library*, arXiv:1910.09336 (2019), doi:10.1145/3372885.3373824 (CPP 2020); the arXiv page was fetched on 2026-10-26 to check title and authors.

[4] H. L. Montgomery, *Minimal theta functions*, Glasgow Math. J. 30 (1988) 75–85. Cited only through [1] (second-hand); the original was not opened. Context: the static optimality of the triangular lattice for Gaussians; nothing here depends on it.